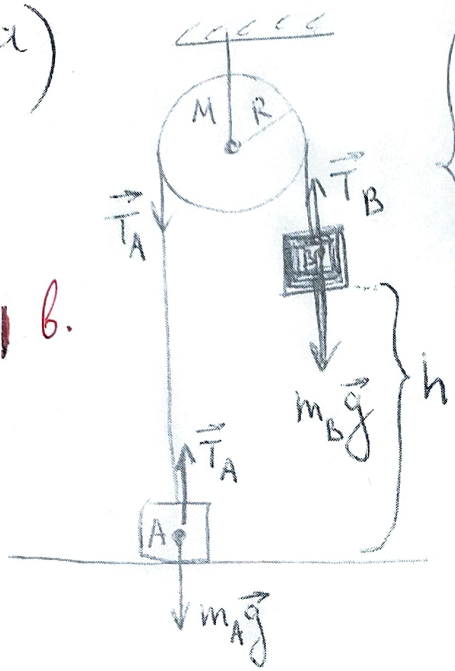
1 - masalaning yechimi. Ip cho'zilmas (ideal) bo'lgani uchun $|\vec{a}_A| = |\vec{a}_B| = a$ 0,5 b.

a)



1 b.

$$\begin{cases} m_B g - T_B = m_B a & (1)\ 1\text{ b.} \\ T_A - m_A g = m_A a & (2)\ 1\text{ b.} \\ (T_B - T_A)R = \frac{1}{2} M R^2 \varepsilon & (3)\ 1{,}5\text{ b.} \end{cases}$$

$$\varepsilon = \frac{a}{R}$$

(1) dan $T_B = m_B g - m_B a$ (4)

(2) dan $T_A = m_A a + m_A g$ (5)

(4) va (5) ni (3) ga qo'yamiz

$$(m_B g - m_B a - m_A a - m_A g)R = \frac{1}{2} M R^2 \frac{a}{R}$$

$$m_B g - m_B a - m_A a - m_A g = \frac{1}{2} M a$$

$$(m_B - m_A) g = \left(m_A + m_B + \frac{1}{2} M\right) a$$

$$a = \frac{m_B - m_A}{m_A + m_B + \frac{1}{2} M} \cdot g \quad (6)\ 1\text{ b.}$$

b) $h = \frac{at^2}{2} \Rightarrow$

$$t = \sqrt{\frac{2h}{a}} = \sqrt{\frac{2h}{g}} \cdot \sqrt{\frac{m_A + m_B + M/2}{m_B - m_A}} \quad (7)\ 1\text{ b.}$$

c) $v_B = a \cdot t = [(6)\cdot(7)] = \sqrt{2gh} \cdot \sqrt{\frac{m_B - m_A}{m_A + m_B + M/2}}$ 1 b.

d) $\varepsilon = \frac{a}{R}$

$\varphi = \frac{\varepsilon t^2}{2} = \frac{at^2}{2R}$

$$N = \frac{\varphi}{2\pi} = \frac{at^2}{4\pi R} = \frac{m_B - m_A}{2m_A + 2m_B + M} \cdot \frac{gt^2}{2R}$$

2 b.

$N = \frac{h}{2\pi R}$ Bu natijani ipning sirpanmasligi shartidan ham hosil qilsa bo'ladi.

2-qism yechimi

1-marta dam berilganda (porshen bosilganida)

$$P_0(V+V_0)^\gamma = P_1 V^\gamma \quad (1)$$

ya'ni, V_0+V hajm V gacha siqiladi, P_0 bosim P_1 gacha ortadi.

$$P_1 = P_0\left(\frac{V+V_0}{V}\right)^\gamma = P_0\left(1+\frac{V_0}{V}\right)^\gamma \quad (1)$$

2-marta dam berilishidan oldin nasos kamerasi havoga to'ldirilayotganda shlang ~~kamerasi~~ klapani yopiladi va ballon ichidagi bosim P_1 qiymatni saqlaydi. Nasos kamerasi yana havoga to'lganida esa uning ichidagi P_0 atmosfera bosimi klapanni ochish uchun yetarli bo'lmaydi. Porshenga 2-marta dam ~~berishdan~~ berilishidan nasos kamerasi ichidagi bosim P_1 ga tenglashguncha klapan ochilmaydi va kamera hajmi V_1 qiymatga erishganida bosim P_1 ga tenglashadi. V_1 ni topamiz:

(1) $\boxed{P_0 V_0^\gamma = P_1 V_1^\gamma}$ $\qquad$ $\boxed{V_1 = \left(\frac{P_0}{P_1}\right)^{1/\gamma}\cdot V_0 = \frac{V_0}{1+\frac{V_0}{V}}}$ (1)

Klapan ochilganidan keyin esa 2-marta ~~dam~~ berishda ballonga havo kirib, bosim P_2 ga teng bo'ladi. P_2 ni topamiz:

$$P_1(V_1+V)^\gamma = P_2 V^\gamma \quad (1)$$

$$P_2 = P_1\left(1+\frac{V_1}{V}\right)^\gamma = P_1\left(1+\frac{V_0}{V+V_0}\right)^\gamma = P_0\left(1+\frac{V_0}{V}\right)^\gamma\left(1+\frac{V_0}{V+V_0}\right)^\gamma \quad (0{,}5)$$

$$\boxed{P_2 = P_0\left(1+\frac{2V_0}{V}\right)^\gamma} \quad (1)$$

Shunday qilib, bu mulohazalarni davom etgan holda, P_N rekkurent munosabatni topamiz (matematik induksiya usuli).

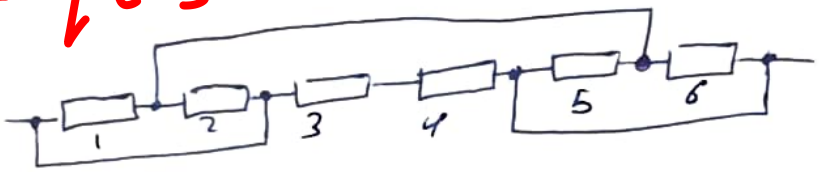
$$\boxed{P_N = P_0\left(1+N\frac{V_0}{V}\right)^\gamma} \quad (1)$$

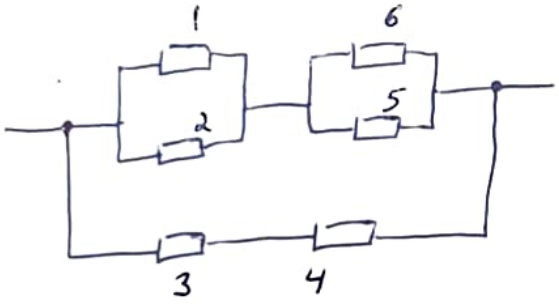
① $P_N = n P_0$ shartdan n ni topamiz:

$$n P_0 = P_0 \left(1 + N \frac{V_0}{V}\right)^{\gamma} \quad (1)$$

$$N = \frac{V}{V_0}\left(n^{1/\gamma} - 1\right) \quad (0,5)$$

3-qism

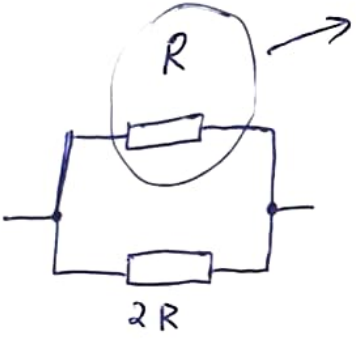




Shu, yoki alternativ holga soddalashtirishga: 6 ball

hato bo'lsa
- 1; 2 parallel va 6,5 parallel: 2 ball
- Rezistordan tok o'tmaydi deb olsa: 0 ball butun masalaga

4-ta rezistorni 1-ta R ga o'zgartirish: 2 ball

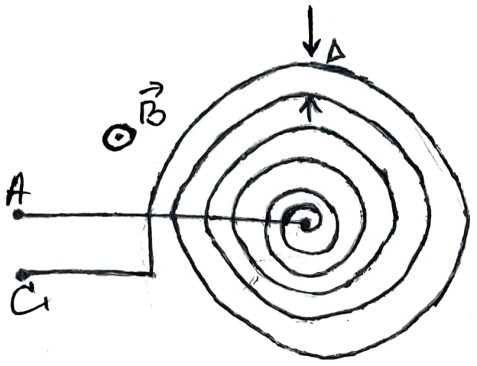


$$R_{um} = \frac{2R \cdot R}{2R+R} = \frac{2}{3}R >$$

Javob: $R_{um} = \frac{2}{3}\Omega$.

To'g'ri javobga: 2 ball

4-qism: (10 ball).



R

$n_1 \ (n >> 1)$

$B = B_0 \sin \omega t$

$\mathcal{E} = ?$ (AC)

Spiralning ikkita uchi, A va C nuqtalar orasidagi induksiyon EYUK har bir xalqaning alohida hosil qilgan induksiyon EYUK'lari yig'indisiga teng.

$$\mathcal{E} = \sum_{i=1}^{n} \mathcal{E}_i \rightarrow (3)$$

$$\mathcal{E}_i = \frac{d\Phi}{dt} = \frac{dB}{dt} \cdot S_i \rightarrow (1)$$

$$\mathcal{E}_i = B_0 \omega \cos \omega t \cdot S_i \rightarrow (1)$$

$$\mathcal{E} = \sum_{i=1}^{n} B_0 \omega \cos \omega t \, S_i$$

$$\mathcal{E} = B_0 \omega \cos \omega t \sum_{i=1}^{n} S_i$$

$$\underbrace{S = \pi R^2}_{0,6}; \qquad \underbrace{S_i = \pi \Delta^2 i^2}_{1,4}$$

$$\mathcal{E} = B_0 \omega \cos \omega t \cdot \pi \Delta^2 \sum_{i=1}^{n} i^2 \rightarrow (2)$$

$$\sum_{i=1}^{n} i^2 = 1^2 + 2^2 + \ldots + n^2 = \frac{n \cdot (n+1)(2n+1)}{6} \rightarrow (0,5)$$

$$n >> 1 \Rightarrow \sum_{i=1}^{n} i^2 = \frac{n^3}{3} \rightarrow (0,5)$$

$$\mathcal{E} = B_0 \omega \cos \omega t \, \pi \Delta^2 \cdot \frac{n^3}{3}; \qquad R = \Delta \cdot n \Rightarrow$$

$$\Rightarrow \boxed{\mathcal{E} = \frac{\pi n B_0 \omega R^2 \cos \omega t}{3}}$$

Ps: Masala boshqa usulda ishlangan bo'lsa va idea to'g'ri hamda javob to'g'ri chiqqan holatda to'liq 10 ball beriladi!

5-masala:

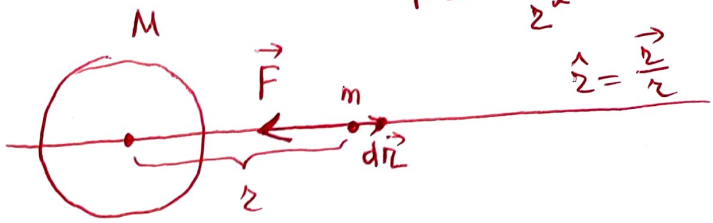
Foton energiyasi $E = h\nu + U$. (1)

U – fotonning potensial energiyasi.

$$dU = -\vec{F}\cdot d\vec{z} = G\frac{mM}{z^2}\cdot dz \quad (2)$$

(chunki $\vec{F}$ va $d\vec{r}$ orasidagi burchak 180° ga teng).

$$\vec{F} = -\frac{GmM}{z^2}\cdot\hat{z} \quad (1,5)$$



$$\hat{z} = \frac{\vec{z}}{z}$$

$dE = 0$; $d(h\nu + U) = 0$; $h d\nu + dU = 0$; $h d\nu + G\frac{mM}{z^2}dz = 0$.

Foton massasi bu modelda $m = \frac{h\nu}{c^2}$ (1,5) bilan aniqlanadi:

$$h d\nu + G\frac{M}{z^2}\cdot\frac{h\nu}{c^2}\cdot dr = 0; \qquad \int_{\nu_1}^{\nu_2}\frac{d\nu}{\nu} = -G\frac{M}{c^2}\int_R^\infty\frac{dz}{z^2}$$

$$\ln\nu\Big|_{\nu_1}^{\nu_2} = G\frac{M}{c^2}\cdot\frac{1}{z}\Big|_R^\infty; \qquad \ln\frac{\nu_2}{\nu_1} = -G\frac{M}{Rc^2}$$

$$\frac{\nu_2}{\nu_1} = e^{-\frac{GM}{Rc^2}} \approx 1 - \frac{GM}{Rc^2} \quad (1); \qquad \nu_2 - \nu_1 = \Delta\nu$$

$$\frac{\nu_2}{\nu_1} = \frac{\nu_1 + \Delta\nu}{\nu_1} = 1 + \frac{\Delta\nu}{\nu_1} \approx 1 - \frac{GM}{Rc^2} \Rightarrow \boxed{\frac{M}{R} \approx -\frac{c^2}{G}\frac{\Delta\nu}{\nu_1}} \quad (2)$$